

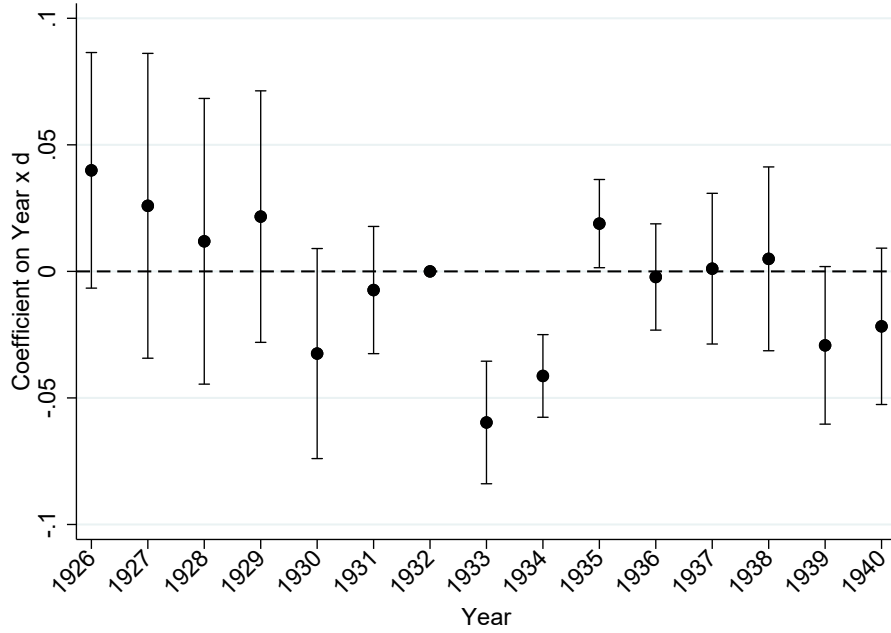
## Response to Referee R6

We sincerely thank you for your time, effort, and valuable feedback. Your detailed comments and constructive suggestions have significantly improved the paper. We have carefully considered each point and prepared a thorough, point-by-point response. Below, we respond to each comment individually; for ease of reading, the original comments are italicized and displayed in red. Before addressing each comment, we begin with a brief summary of the major revisions.

- **Extended sample.** Following R1’s and R6’s suggestions, we extended our panel data coverage from seven years (1930–1936) to fifteen years (1926–1940). This extension serves two primary purposes:
  1. Examining additional pre-treatment years (1926–1932 rather than just 1930–1932) enables a more rigorous evaluation of the parallel trends assumption underlying our difference-in-differences design.
  2. The recession of 1937–1938 provides a natural placebo test: if our gold-clause exposure measure captured something other than the mechanism we propose—such as general sensitivity to economic downturns—we would expect it to affect investment during this later period of distress as well.

Figure R.I plots the year-by-year interaction coefficients from our main investment regression over the extended sample. The figure confirms the absence of pre-trends before 1933, sharp negative effects during the 1933–1934 litigation period, and no effect during the 1937–1938 recession.

- **Data reconstruction.** While extending the sample, we identified that some scans purchased from Mergent were of poor quality, leading to firm omissions and data entry errors. To address these concerns, we undertook a complete data reconstruction:
  1. We scanned, in high resolution, the Moody’s Industrial Manuals for all sixteen years, ensuring complete legibility of all pages.



**Figure R.I.** Gold clause effect on investment by year

*Notes.* This figure plots point estimates and 95% confidence bands from a regression of net investment on  $Q$  and  $\tilde{d}$  interacted with year indicators, including firm and year fixed effects.

2. We recollected the entire dataset using a double-blind procedure in which two independent research assistants collected data separately, with discrepancies resolved by a third party.
3. We implemented extensive quality control checks to identify and correct any remaining issues.

This reconstruction process required nearly two years to complete but was essential for ensuring the reliability of our findings.

- **Gold clause exposure measure.** Following R1's suggestion, we replaced total debt with long-term liabilities in the denominator of our gold exposure measure,  $d_{j,t}$ , now defined as the ratio of gold-clause bonds to long-term liabilities (corporate bonds, preferred shares, and

bank loans) for firm  $j$  at time  $t$ :

$$d_{j,t} = \frac{\text{Gold-denominated long-term liabilities of firm } j \text{ in year } t}{\text{Long-term liabilities of firm } j \text{ in year } t}.$$

- **Addressing endogeneity concerns.** Following R6’s suggestion, we fix each firm’s gold exposure at its 1930 value for all subsequent years to address potential endogeneity from strategic adjustments following Britain’s departure from the gold standard in September 1931. Formally:

$$\tilde{d}_{j,t} = \begin{cases} d_{j,t}, & \text{if } t \leq 1930 \\ d_{j,1930}, & \text{if } t > 1930 \end{cases}.$$

This measure,  $\tilde{d}$ , is our main explanatory variable throughout the revised analysis.

- **Empirical design.** Following R1’s, R2’s, and R6’s suggestions, we now use the full 1925–1940 panel and estimate the effects of gold clause exposure and its reversal in a single regression.
- **Debt overhang framing.** Following R1’s and R2’s suggestions, we have substantially revised the paper’s conceptual framing. We no longer use the term “leverage risk” in the paper title or text and instead interpret the events of 1933–1934 through the lens of dynamic debt overhang theory (Hennessy, 2004). This framing more precisely captures the economic mechanism linking gold clause exposure to investment decisions.
- **Firm-level uncertainty literature.** Following R2’s suggestion, we have repositioned the paper within the firm-level uncertainty literature (Bloom, 2009; Baker et al., 2016; Hassan et al., 2019). The uncertainty surrounding the gold clause represents a policy shock of the type studied extensively in this growing literature. This reframing highlights that our gold clause exposure measure,  $\tilde{d}_{j,t}$ , captures firm-level variation in exposure to policy uncertainty—derived from balance sheet data rather than earnings call transcripts as in (Hassan et al., 2019). We are grateful to R2 for this suggestion, as it clarifies how our historical setting illustrates the real effects of policy uncertainty during a pivotal episode.

### *1. Gold clause or sensitivity to leverage*

*Ideally, to argue that gold-denominated debt is driving the results, one could want to contrast two firms with similar levels of interest-bearing debt but with variation in the fraction of that debt that is gold-denominated. The authors' main measure relates bonds outstanding with gold clauses relative to total liabilities, and controls for book and market leverage, measured by liabilities to assets. What this paper does not do, and what I think they may not be able to do, is to control for interest-bearing debt to total liabilities, or instead relate gold-denominated debt to interest-bearing debt as their main variable. The reason why I think they may not be able to do it is because the vast majority of bonds were indexed, and bank loans were tiny. What this means is that it is hard to know whether the indexing or the level of interest bearing debt are truly driving the results. One can think of reasons why variation in leverage may matter differentially over time independent of indexation. For example, bond markets were especially depressed in 1933 and 1934, but recovered substantially in 1935 and 1936. This is also when the second New Deal reactivated the economy, albeit temporarily. So it would be important to better disentangle gold-indexed bonds versus other bonds, as described above (and the summary statistics for these two should be listed in the paper). If this cannot be done, the paper needs to be more upfront about this important limitation.*

Your impression is correct. In 1932, about 97% of the corporate bonds in our sample had the gold clause and these bonds represent 99% of total bond amount outstanding. In our original submission, we had that information in a footnote, but we agree that such an important piece of information should be featured more prominently in the paper.

To clarify this point, we now mention it on the first page of the introduction: “When the United States effectively abandoned the gold standard in April 1933, the dollar quickly depreciated against gold, and the existence of these clauses posed a serious threat to most corporate bond issuers: around 97% of outstanding corporate bonds were subject to a gold clause.” We hope that explicitly stating the exact number early in the introduction leaves no room for ambiguity. We have also added details on the number of corporate bonds that contained the gold clause versus those that did not in Table 2 of the revised manuscript.

Motivated by your comment, as well as a related suggestion from another referee, we have revised the definition of our exposure measure  $d$ . In the original submission, the denominator was total liabilities (which, as you noted, includes non-interest-bearing debt). In the revision, the denominator is the sum of corporate bonds, preferred shares, and bank debt—that is, all long-term obligations involving fixed contractual payments:

$$d_{j,t} = \frac{\text{Gold-denominated debt}_{j,t}}{\text{Corporate bonds}_{j,t} + \text{Preferred shares}_{j,t} + \text{Bank debt}_{j,t}}.$$

While corporate bonds and preferred shares are not perfect substitutes, they are similar in that they are both claims to future cash flows that are subject to fixed contractual payments. This definition ensures that variation in  $d$  more closely reflects differences in the *composition* of fixed-payment claims, rather than differences in its *level*.

In sum, we have made the 97% figure more prominent in the introduction, provided detailed bond statistics in Table 2, and revised the definition of  $d$  to better isolate variation in gold indexation from variation in the level of interest-bearing debt.

## *2. Broader macro story, of just large firms?*

*Only the largest firms in the economy issued bonds. In fact, more than ¼ but less than ½ of the firms in the sample had gold-denominated bonds. Was this a more general issue in the economy at large, for firms or households? If yes, it would be great to provide some descriptive evidence. If not, it shapes how we see the contribution of the paper to our understanding of the Depression. Yes, understanding large firm behavior is important, but the macro estimates in the latest section of the paper could not be extrapolated to the rest of the economy, making the effects of this channel on the very sizable contraction in private non-residential fixed investment much smaller.*

We thank you very much for this insightful comment. You are correct to point out that, in our original submission, we were not adequately discussing the relevance of our findings to the broader economy. In the revised manuscript, we address this issue in three places: (i) the “Historical background” section, (ii) the “Aggregation” section, and (iii) the “Conclusion.”

First, we now discuss in the “Historical background” section how the issue of gold clauses was understood to extend beyond industrial firms at the time. We quote President Hoover’s 1932 campaign speech in which he warned that “A considerable part of farm mortgages, most of our industrial and all of our Government, most of our State and municipal bonds, and most other long-term obligations are written as payable in gold.” We also include his warning to farmers that, if the US had gone off gold, they “would have sold their produce for depreciated currency... but you would have to pay a premium on such of your debts as are written in gold.” Given that agriculture employed nearly a quarter of the workforce at the onset of the Depression, the prevalence of gold clauses in farm mortgages underscores the broader economic relevance of the abrogation.

Second, in the “Aggregation” section, we now explicitly interpret our results as providing a lower bound for the total effect of gold clause abrogation on private investment:

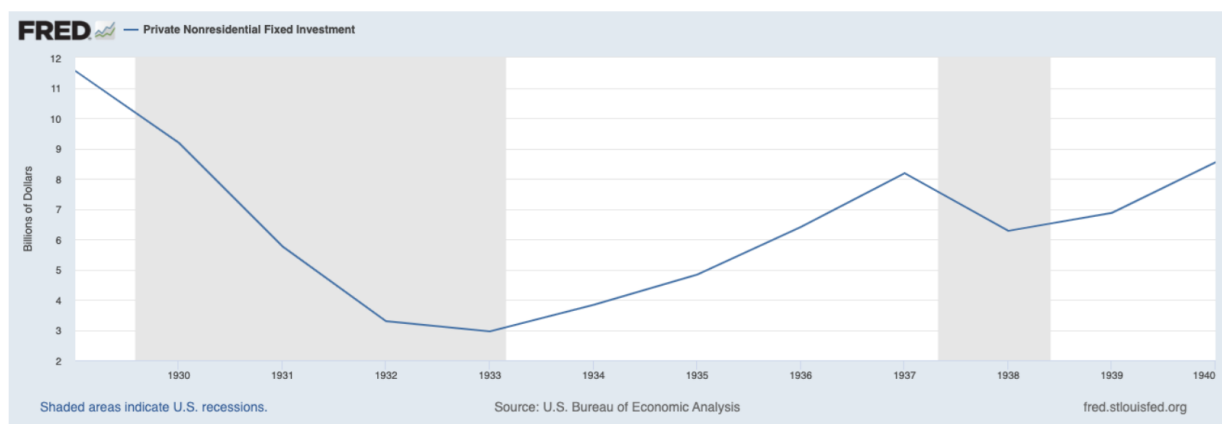
Finally, we interpret our aggregation results as providing a lower bound for the total effect of gold clause abrogation on private investment. As previously discussed, gold clauses were prevalent in other private contracts, including farm mortgages, public utility bonds, and railroad securities. The coverage of our data does not allow us to study the effects of the abrogation on these other borrowers, but our results suggest that the abrogation may have had economically significant effects on these sectors as well, which is an interesting avenue for future research.

Third, the “Conclusion” now identifies other sectors where the gold clause was prevalent as promising avenues for future research:

While this paper focuses on the impact of gold clause abrogation on large industrial firms, examining its effects on other sectors of the economy represents a promising avenue for future research. Evidence suggests that gold clauses were prevalent in farm mortgages, making the agricultural sector a particularly important case for investigation. Given that agriculture employed nearly a quarter of the workforce at the onset of the Depression, understanding how the abrogation affected farmers’ debt burdens, investment decisions, and recovery patterns could provide crucial insights into the pol-

ity's broader economic impact. Similarly, public utilities and railroads—major issuers of gold-denominated corporate bonds and critical components of economic infrastructure—warrant further attention.

An inspection of the Moody's Steam Railroads Manuals and the Moody's Public Utilities Manuals reveals that firms in these industries were large issuers of gold-denominated corporate bonds. We have excluded these firms from our analysis, because they were heavily regulated and very few were publicly listed which makes it difficult to measure their average  $Q$  and implement standard investment regressions. Nonetheless, we believe that examining the impact of the gold clause on these industries—as well as on agricultural investment—represents a promising avenue for future research.



**Figure R.II.** Private non-residential investment

*As shown above, most of the contraction in private fixed investment took place before 1932. Again, though the allocation within firms is significant, stating that "Taken together, our partial equilibrium aggregation exercise suggests that the uncertainty surrounding the abrogation of gold clauses led to a steep decline in public firms' investment and, therefore, likely contributed to the slow speed of economic recovery from the Great Depression. We find that leverage risk led public firms to delay investment in fixed capital for two years. This finding complements existing explanations of the delayed recovery based on the disruption of credit intermediation, which is more applicable to smaller firms. " seems much too strong, and the last sentence goes against the findings of Benmelech*

*et al (2019), who show that the disruption in intermediation was quantitatively important for large firms early in the crisis.*

You are right to point out that our original claim was overstated. The passage could be read as suggesting that the gold clause abrogation caused the initial contraction in investment, which clearly contradicts the empirical evidence shown in the figure you provided—most of the decline in private fixed investment occurred before 1932, prior to the abrogation. This is not the idea we intended to convey. In the revised manuscript, we have substantially toned down the implications of the aggregation exercise. The revised text in Section 5.6 now states:

“Taken together, the partial-equilibrium aggregation results highlight the significant role of gold clause uncertainty in explaining the weak investment recovery of public firms in 1933 and 1934.”

This formulation emphasizes that our findings pertain to the *recovery period* (1933–1934) rather than to the initial contraction, and explicitly limits the scope of our claims to public firms. We have also removed language that could be interpreted as suggesting gold clause uncertainty was the primary driver of aggregate investment dynamics.

Moreover, we recognize that our original wording could be read as dismissive of the importance of the credit intermediation channel for large firms. As you correctly noted, this would be inadequate in light of the findings in Benmelech, Frydman, and Papanikolaou (2019), who document that disruptions in intermediation and bond market dislocations were quantitatively important for large firms in the early years of the crisis. The revised Conclusion now frames our contribution more carefully:

“Taken together, the evidence shows that policy-driven legal uncertainty on the liability side can depress investment at scale, even in the absence of realized changes in leverage. By isolating a plausibly exogenous episode that shifts expected liabilities, we provide evidence consistent with dynamic debt overhang and quantify its aggregate relevance.”

This revised framing avoids any suggestion that our results supplant or contradict the credit inter-



mediation literature, and instead positions our findings as documenting a distinct mechanism—legal uncertainty affecting the liability side—that operated alongside credit frictions during the recovery period.

*The statement that "large firms, with their ample cash holdings, were mostly insulated from the distress of the financial intermediation sector (see Lutz (1945), Hunter (1982), Bernanke (1983), and Calomiris and Mason (2003))" is also highly misleading because these papers are either not based on firm-level data or just have information on a handful of firms. Using over 1000 large firms, the cited paper by Benmelech clearly upends this view. This is acknowledged by Bernanke in his 2018 Brookings review article on the financial crisis. Firms needed cash for many reasons, not just investment, so the authors need to be much more cautious in their descriptions on constraints throughout.*

Thank you for directing us to Bernanke’s Brookings Review article, which we were not aware of. After reviewing it, we recognize that our original assertion—that “large firms, with their vast cash holdings, were mostly insulated from the distress of the financial intermediation sector”—was overstated. We have revised the paper accordingly and now acknowledge in Section 5.1 that Benmelech, Frydman, and Papanikolaou (2019) “show that disruptions in intermediation and bond market dislocations were key drivers of the investment contraction in the early years of the Great Depression, even for large firms.”

Finally, we acknowledge that our discussion of financial constraints throughout the paper required greater nuance. In the revised manuscript, we have systematically adjusted our claims in each instance where financial constraints are discussed to better reflect the current state of the literature and avoid overstating our conclusions.

### *3. Adjustments in $d_i$*

*I understand why the authors would want to keep  $d_i$  constant at the pre-abrogation decision level. But this does not mean that firms should not have strategically adjusted before, if they could. As the authors noted, in September 1931 UK left the gold standard, which was once unthinkable. This triggered a speculative attack on the dollar, under the belief that a devaluation in the US was imminent,*

*and a banking panic in the US: <https://www.federalreserve.gov/boarddocs/speeches/2004/200403022/default.htm>. While the Fed chose to support the Gold Standard, it is quite reasonable to believe that uncertainty about a possible devaluation and what it would mean to borrowers should have been heightened. Especially since legal precedent was that contracts in gold would be upheld, one would expect unconstrained firms run by savvy managers to pay down their gold-denominated debts as much as possible. And to do so before December 1932. Thus, it is unclear one could treat the 1932 numbers as exogenous. Defining treatment as of 1932 is pre gold clause but not pre depression or pre dislocations in debt markets.*

Thank you for raising this important point. You are right that Britain’s departure from the gold standard in September 1931 could have prompted firms to anticipate similar policy changes in the United States, leading them to strategically adjust their gold clause exposure. If only financially unconstrained firms were able to reduce their gold-denominated liabilities prior to the abrogation, the remaining firms with high gold exposure would disproportionately consist of constrained firms, potentially biasing the estimated effect of gold exposure on investment by conflating it with the impact of pre-existing financial constraints. This concern is critical to the validity of our empirical setting, and we have devoted considerable effort to assessing its impact on our results.

We begin by examining whether firms actually adjusted their gold clause exposure in the years preceding the abrogation. Table 2 of the revised manuscript tracks the evolution of gold-denominated liabilities from 1930 to 1935 for a balanced panel of 157 firms with corporate bonds outstanding at the end of 1930. The evidence suggests minimal adjustment prior to 1933.

On the extensive margin, of the 150 firms with at least some gold-denominated bonds at the end of 1930, 144 still had gold-denominated bonds by the end of 1932. The total number of gold-denominated bonds declined only modestly, from 250 to 240 over the same period. On the intensive margin, the average value of gold clause exposure,  $d_{j,t}$ , among exposed firms remained virtually unchanged at 0.57 in 1930 and 0.58 in 1932. The stability of average  $d$  among exposed firms suggests that selection—whereby only unconstrained firms reduced their exposure—was minimal. In contrast, the pace of adjustment accelerated markedly after the abrogation: 36 bonds were redeemed in 1933 and 15 in 1934.

We were initially surprised by how little adjustment occurred prior to 1933, which led us to investigate the historical context more thoroughly. We have expanded Section 3 (Historical Background) to provide a fuller account of why firm managers may not have anticipated the abrogation.

On the political front, President Hoover remained a staunch defender of the gold standard throughout his presidency, describing it as “a little short of a sacred formula” (Eichengreen and Temin, 2000). Even during the 1932 presidential campaign, Hoover attacked Roosevelt for allegedly proposing fiat money (Edwards, 2018), while Roosevelt vigorously denied any intention to abandon gold, pointing to the gold clauses in Treasury bonds as evidence of the government’s commitment.

On the economic front, even if some managers perceived the risk, market conditions may have made debt restructuring prohibitively expensive. As Figure IA.1 in the Internet Appendix shows, many industrial bonds traded at steep discounts to par value in 1931, which implies that most call options were deeply out-of-the-money. Open-market repurchases would have been similarly costly, as buying large quantities of bonds would likely have moved prices against the firms.

Finally, the exchange rate evidence corroborates the view that the abrogation came as a surprise: as shown in Figure 1 of the revised manuscript, the dollar remained stable against sterling and the franc until the suspension of convertibility in April 1933, at which point it depreciated rapidly.

Nonetheless, following one of your subsequent suggestions, we now construct a modified measure,  $\tilde{d}_{j,t}$ , that fixes each firm’s gold clause exposure at its 1930 level—before Britain’s departure from the gold standard—for all subsequent years:

$$\tilde{d}_{j,t} = \begin{cases} d_{j,t}, & \text{if } t \leq 1930 \\ d_{j,1930}, & \text{if } t > 1930 \end{cases}.$$

By construction,  $\tilde{d}_{j,t}$  is immune to any strategic adjustments firms may have made after 1930. We use  $\tilde{d}_{j,t}$  as our primary exposure measure throughout all regressions in the revised manuscript.

*Similarly, it is unclear to me whether firms were able to pay interest and principal at the devalued or at the pre-abrogation levels from June 1932 to February 1935. If so, the reason why uncertainty may have affected activity is that they may have chosen to massively repay in case the Supreme*

*Court decided against the devaluation. So they paid more to creditors and shareholders. Is that the case? Note that dropping firms with maturing bonds does not help here.*

This is an important point that was not sufficiently clear in our original submission. Between June 1933<sup>1</sup> and February 1935, firms were able to make payments at the devalued dollar rate. That is, a bond with a \$25 semi-annual coupon continued to pay exactly \$25 throughout the litigation period and beyond, rather than an amount adjusted for the dollar’s devaluation against gold. We have clarified this in Section 3 with a footnote stating: “Throughout the litigation surrounding the abrogation of the gold clause, payments on bonds containing the clause were not indexed to gold.” As Edwards (2018) reports, lower courts had already ruled against indexing to gold as early as May 1933. The Joint Resolution then made the abrogation law, and the Supreme Court upheld it in 1935.

You also raise the interesting point that in 1933–1934 firms may have redeemed bonds to protect themselves against the risk that the Supreme Court would rule against abrogation. The evidence in Table 2 is consistent with this view: the number of gold-denominated bonds declined from 250 in 1930 to 212 in 1933 and 189 in 1934. We agree that dropping firms with maturing bonds does not address this concern, since firms could have voluntarily redeemed bonds to reduce exposure. To test whether this redemption channel drives our results, we re-estimate our baseline specification excluding firms that repurchased their bond issues in 1933 and 1934. As reported in Column 4 of Table 3, the interaction coefficients for 1933 and 1934 remain economically and statistically significant (-0.068 and -0.032, respectively). These results indicate that the decline in investment among gold-exposed firms extends beyond those actively managing their debt obligations through redemptions during the litigation period.

We greatly appreciate this point, as it has provided substantial clarity both for us and for readers.

*It would be nice to do validations with earlier ratios (1930 or ideally 1929), as well as to show the evolution of this ratio from 1929 to 1936.*

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<sup>1</sup>We believe you meant June 1933 rather than June 1932.

Following your suggestion here and in Comment 4, we now use  $\tilde{d}_{j,t}$  as our measure of gold clause exposure. By definition,  $\tilde{d}_{j,t}$  fixes each firm’s exposure at its 1930 level for all subsequent years, ensuring that post-1930 adjustments do not affect our treatment variable. Measuring exposure in 1930 places it before both the leverage uncertainty period and Britain’s departure from the gold standard.

We also report the evolution of  $d$  in Table 2 of the revised manuscript. The table shows that the number of firms with positive gold clause bonds was stable until 1933, then declined, while the average and median  $d$  (conditional on having gold clause bonds) remained around 58%. This pattern suggests that adjustments in gold clause exposure occurred primarily on the extensive margin, with some firms exiting entirely rather than gradually reducing their exposure.

*One other historical validation would be to look at the downturn in 1937-38. By then gold denomination should not have mattered. Did interest bearing debt still affect investment choices then?*

This is a valuable point that we can now address with the additional data collected (our original sample ended in 1936). Table 3 shows that in 1937 and 1938 the coefficient on  $\tilde{d}$  is not statistically significant, consistent with the expectation that the impact of gold clauses disappeared after the Supreme Court’s decision.

The absence of effects in 1937 and 1938 is particularly noteworthy because these were also years of economic distress. As you note, by then gold denomination should not have mattered—and indeed, we find that the composition of fixed-payment liabilities (gold-denominated versus non-gold-denominated) no longer affected investment choices. If  $\tilde{d}$  instead captured inherent firm characteristics that made some companies more vulnerable during downturns, we would expect to observe differential investment patterns whenever economic conditions deteriorated.

We interpret these results as suggestive evidence that our 1933–1934 findings reflect the specific impact of gold clause litigation rather than persistent differences in firms’ sensitivity to adverse economic conditions. We highlight this finding in Section 5.2 as an important validation test for our results.

*It would also be helpful to see what happened to the market value of firms by  $d_i$  around the September 1931 event.*

Table R.I reports market performance metrics around Britain’s departure from the gold standard on September 21, 1931. We construct a  $d$ -weighted portfolio in which each firm’s weight is proportional to its gold clause exposure  $d_{j,t}$ , so that firms with higher exposure receive greater weight. Panel A presents daily returns for the market portfolio, the  $d$ -weighted portfolio, and abnormal returns of the  $d$ -weighted portfolio using both CAPM and Fama-French three-factor (FF3F) models for risk adjustment. Panel B presents cumulative returns over the same period, normalized to zero on September 19, 1931—the last trading day before Britain left the gold standard.

| <i>Panel A: Daily Returns</i>                                |         |                         |                 |                 |
|--------------------------------------------------------------|---------|-------------------------|-----------------|-----------------|
| Date                                                         | Market  | $d$ -Weighted Portfolio | Abnormal (CAPM) | Abnormal (FF3F) |
| 9/15/1931                                                    | -0.0129 | -0.0242                 | -0.0127         | -0.0058         |
| 9/16/1931                                                    | -0.0097 | -0.0137                 | -0.0054         | -0.0065         |
| 9/17/1931                                                    | 0.0135  | 0.0081                  | -0.0066         | 0.0066          |
| 9/18/1931                                                    | -0.0445 | -0.0429                 | -0.0001         | -0.0102         |
| 9/19/1931                                                    | -0.0341 | -0.0249                 | 0.0077          | 0.0180          |
| 9/21/1931                                                    | -0.0156 | -0.0325                 | -0.0184         | 0.0070          |
| 9/22/1931                                                    | -0.0106 | -0.0123                 | -0.0031         | 0.0061          |
| 9/23/1931                                                    | 0.0645  | 0.0630                  | -0.0022         | -0.0014         |
| 9/24/1931                                                    | -0.0677 | -0.0263                 | 0.0395          | 0.0117          |
| 9/25/1931                                                    | 0.0160  | -0.0017                 | -0.0188         | 0.0133          |
| 9/26/1931                                                    | -0.0117 | 0.0068                  | 0.0171          | -0.0099         |
| <i>Panel B: Cumulative Returns (normalized to 9/19/1931)</i> |         |                         |                 |                 |
| Date                                                         | Market  | $d$ -Weighted Portfolio | Abnormal (CAPM) | Abnormal (FF3F) |
| 9/15/1931                                                    | 0.0748  | 0.0734                  | 0.0044          | -0.0078         |
| 9/16/1931                                                    | 0.0651  | 0.0597                  | -0.0010         | -0.0144         |
| 9/17/1931                                                    | 0.0786  | 0.0678                  | -0.0076         | -0.0078         |
| 9/18/1931                                                    | 0.0341  | 0.0249                  | -0.0077         | -0.0180         |
| 9/19/1931                                                    | 0.0000  | 0.0000                  | 0.0000          | 0.0000          |
| 9/21/1931                                                    | -0.0156 | -0.0325                 | -0.0184         | 0.0070          |
| 9/22/1931                                                    | -0.0262 | -0.0448                 | -0.0214         | 0.0131          |
| 9/23/1931                                                    | 0.0383  | 0.0181                  | -0.0237         | 0.0117          |
| 9/24/1931                                                    | -0.0294 | -0.0082                 | 0.0158          | 0.0234          |
| 9/25/1931                                                    | -0.0134 | -0.0099                 | -0.0030         | 0.0367          |
| 9/26/1931                                                    | -0.0251 | -0.0031                 | 0.0141          | 0.0268          |

**Table R.I** Market returns around UK’s departure from gold standard

Panel A shows that the market reacted negatively to Britain’s departure from the gold standard on September 21, 1931, with a return of  $-1.6\%$ . Consistent with the view that firms with gold-denominated debt had more to lose if the US followed suit, the  $d$ -weighted portfolio declined by  $-3.3\%$  that day—more than twice the market decline. The CAPM abnormal return of  $-1.8\%$  supports the same interpretation. However, the FF3F abnormal return on that day was slightly positive ( $+0.7\%$ ), suggesting that part of the underperformance may be attributable to size and value factor exposures rather than gold clause exposure per se.

However, Panel B reveals that this underperformance was short-lived. Starting from zero on September 19, the cumulative return of the  $d$ -weighted portfolio dropped to  $-3.3\%$  on September 21, versus  $-1.6\%$  for the market. But by September 24, the  $d$ -weighted portfolio ( $-0.8\%$ ) had largely recovered and outperformed the market ( $-2.9\%$ ). By the end of our window on September 26, the  $d$ -weighted portfolio had outperformed the market by 2.2 percentage points ( $-0.3\%$  versus  $-2.5\%$ ). The cumulative CAPM abnormal return, which reached  $-1.8\%$  on September 21, had fully reversed by September 26 and turned positive ( $+1.4\%$ ). The cumulative FF3F abnormal return was positive throughout the post-event period, reaching  $+2.7\%$  by September 26.

This evidence is consistent with the view that while markets may have initially perceived gold-exposed firms as somewhat vulnerable to a potential US departure from the gold standard, concerns appear to have dissipated within days. As discussed in Section 3, the Federal Reserve responded to Britain’s departure by raising its discount rate, which swiftly stemmed gold outflows and signaled the government’s commitment to defending the standard. Moreover, as documented earlier in this response, firms did not subsequently adjust their gold clause exposure despite having nearly two years to do so before the abrogation. This inaction, combined with the exchange rate stability shown in Figure 1 of the revised manuscript and the reversal of abnormal returns documented in Table R.I, is suggestive—though not conclusive—that initial concerns were short-lived.

*If gold-denominated debt cannot be differentiated from bonds, then it is possible that the effects are instead driven by the constraints in making high interest payments during tough years (1933-34) relative to recovery years? Thus, there is a conflict of interest between creditors and shareholders,*

*but it is not driven by the gold clause.*

This is a legitimate concern. In our original submission, gold-denominated debt could not be readily distinguished from bonds more generally, making it difficult to separate the effect of gold clauses from the effect of high fixed payments.

In response to your concern and a related suggestion from R1's suggestion, we have revised our definition of  $d$ . As now defined in equation (1), both the numerator (gold-denominated corporate bonds) and the denominator (corporate bonds plus preferred stock) consist of securities requiring fixed payments. Importantly, preferred dividend rates were typically higher than bond coupon rates, consistent with preferred stock's subordinated claim on firm cash flows. Thus, if the effects were driven by constraints from making high fixed payments during tough years, we would not expect variation in  $d$ —the share of fixed-payment liabilities that is gold-denominated—to predict investment.

We acknowledge that preferred dividends are optional rather than mandatory. However, in our review of the Moody's Manuals, we did not find many instances where preferred dividends were suspended in 1933 and 1934. Moreover, we found that exposed firms increased their regular (common) dividends more than unexposed firms during the litigation period, providing indirect evidence that preferred payments were maintained: to pay common dividends, firms must first be current on their preferred obligations.

This evidence is of course not definitive, but it is consistent with the view that shareholders were extracting value amid gold clause uncertainty rather than firms facing binding cash flow constraints from fixed payments.

#### *4. Differences in trends and panel structure*

*Another concern with defining treatment in 1932 is that more profitable firms in 1931-32 have less gold-denominated debt. It is possible that firms that were more recession proof paid their debt disproportionately, and also did better during tough years in 1933 and 1934. Ruling this omitted variable is challenging.*

Thank you for raising this concern, which is distinct from, yet related to, Comment 3. Whereas



Comment 3 focused on whether firms strategically anticipated the abrogation, this comment raises the possibility that firms with lower gold exposure in 1931–32 were simply more profitable or “recession-proof,” and that these characteristics—rather than gold clause exposure—explain their stronger performance in 1933–34.

In the revision, we look at this concern in several ways. First, we now measure  $\tilde{d}$  in 1930 throughout the revised manuscript. This timing reduces the scope for reverse causality whereby 1931–32 profitability determines both gold exposure at **teh** end of 1932 and investment in subsequent years.

Second, as documented in our response to Comment 3, we find minimal adjustment in gold clause exposure prior to 1933. The evidence in Table 2 shows that 144 of 150 firms with gold-denominated bonds in 1930 still had such bonds in 1932, and the average  $d$  among exposed firms remained virtually unchanged (0.57 in 1930, 0.58 in 1932). This stability suggests that the hypothesized selection—whereby recession-proof firms disproportionately paid down gold debt—was limited in practice.

Third, we directly test whether the firms that adjusted the gold content of their liabilities drive our results. As reported in Column 4 of Table 3, we re-estimate our baseline specification excluding firms that redeemed bond issues during 1933–1934, years where we find significant evidence of adjustment. The interaction coefficients for 1933 and 1934 remain economically and statistically significant, indicating that our findings extend beyond firms actively managing their debt obligations.

We discuss these robustness checks in Section 5.2 of the revised manuscript.

*It is also hard within the structure of the data to argue that there are no pre-trends. I do not understand why the information is treated over two-year intervals, as opposed to using a panel structure. This would allow authors to show year-to-year estimates of the effects of  $d_i$ , ideally measured in 1930, over time, and show the specific years in which investment changes. It would also simplify the comparison over time. Similar magnitudes of the main coefficient over the two main periods with reversing signs are harder to contrast when the entire point is that investment declined over the first period.*

We fully agree that this approach is preferable and have restructured the analysis accordingly. In the revised manuscript, we implement an event-study specification that interacts  $\tilde{d}$  (measured in 1930) with year indicators for each year from 1926 to 1940, omitting 1932 as the reference year. This allows us to trace out the year-by-year relationship between gold clause exposure and investment in a single regression framework.

Table 3 of the manuscript reports the coefficient estimates from this specification. Figure 3 plots the interaction coefficients along with 95% confidence intervals, providing a visual assessment of pre-trends and treatment effects. The results show: (i) no statistically significant relationship between  $\tilde{d}$  and investment in the pre-treatment years (1926–1932), supporting the parallel trends assumption; (ii) negative and statistically significant coefficients in 1933 and 1934, the years of gold clause litigation; and (iii) a return to statistically insignificant coefficients after the Supreme Court’s 1935 decision. This pattern—no effect before, negative effect during, no effect after—is consistent with gold clause uncertainty affecting investment during the litigation period, though we acknowledge that alternative explanations cannot be fully ruled out. We also note that the 1935 interaction coefficient is positive and marginally significant, suggesting a partial rebound in investment by exposed firms following the Supreme Court’s resolution of the gold clause question.

The year-by-year structure also reveals that the investment decline was concentrated in 1933, with a smaller (though still significant) effect in 1934.

*It is also hard to understand the findings when the panel is not balanced, because firms may be dropping in and out of the sample for endogenous reasons. Say that only firms with  $di=0$  go bankrupt. Would these have investment rate of minus infinity? A cleaner way would be to at least run a robustness check with a balanced panel.*

We agree that endogenous attrition is a valid concern and conduct robustness checks with balanced panels. Internet Appendix Table IA.12 reports the results. Columns 2–4 repeat our baseline investment regression restricting the sample to firms with uninterrupted data for three progressively longer windows: 1930–1936 (7 years), 1929–1940 (12 years), and 1926–1940 (15 years).

Overall, the results are robust to requiring balanced panels. For the 1930–1936 balanced panel,

the interaction coefficients for 1933 and 1934 are  $-0.035$  and  $-0.033$ , respectively, both statistically significant at the 1% level. For the 1929–1940 balanced panel, the coefficients are  $-0.025$  and  $-0.029$ , again both significant at the 1% level. Even for the most demanding specification—a fully balanced panel spanning 15 years from 1926 to 1940—the 1933 coefficient remains negative and significant ( $-0.032$ ), though the 1934 coefficient loses significance. This attenuation is not surprising given the substantial reduction in sample size (from over 6,000 observations to 3,630) required to maintain a balanced panel over such a long horizon.

Importantly, in all balanced panel specifications, the interaction coefficients for years outside the litigation period remain statistically insignificant, supporting the parallel trends assumption.

## 5. *Other comments*

### *a. Why is investment rate trimmed and all other variables winsorized? Can you be consistent?*

Thanks for pointing this out. We now apply a consistent approach: all variables are winsorized at the 0.5% and 99.5% levels within each year throughout the paper. This is now documented in the notes of all tables presenting regression results, which state: “All variables are winsorized at the 0.5% and 99.5% levels within each year.”

### *b. How can payout be negative for some firms?*

Our measure of payout includes dividend payment and net share repurchases (share repurchases minus share issuance). That is, if the net value of the shares a firm issued surpasses the firm’s dividend payment in a given year, its payout is negative. Having said that, we now report results separately for payout components in Table 4. Column 1 reports results for total payout, while columns 2 and 3 decompose the effect into dividends and net repurchases. The results show that the payout effect is driven entirely by cash dividends (column 2), which are always non-negative. The dividend coefficients for 1933 and 1934 are 0.028 and 0.036, both statistically significant, while the net repurchase coefficients (column 3) are small

and insignificant. This pattern is consistent with the observation that before 1982, dividend payments were the primary driver of payout policy.

*c. Why is payout defined over fixed assets? This is certainly not standard.*

Following your suggestion, we now use a more standard definition: payouts (including dividends and net repurchases) are normalized by the book value of common stock in 1930. This is now stated in the notes to Table 4 and defined in equation (4) of Section 5.3.

In response to a suggestion from R1, we also experimented with alternative definitions of dividends and report the results in Internet Appendix Table IA.15. This table examines eight alternative specifications: (1) dividends normalized by 1930 fixed capital for firms that paid positive dividends in the previous year, (2) the same for firms that paid zero dividends, (3-4) similar splits based on 1932 dividend status, (5) annual dividend growth, (6) dividends divided by lagged book equity, (7) dividends divided by 1930 market capitalization, and (8) dividends as a share of net income for profitable firms. The results are generally consistent across specifications, with positive point estimates for 1933 and 1934 that are statistically significant in most cases. The sample size varies across specifications, particularly for column (8), where requiring positive net income that exceeds dividends substantially reduces observations.

*d. Why is the sample restricted to publicly traded firms? Prices are almost not necessary, other than for the market leverage control, and this comes at the cost of losing about half the firms in the Moodys manuals.*

Besides putting convenient bounds on the scope of our data collection, focusing on publicly traded firms is essential for our conceptual framework. The revised manuscript relies on the investment- $Q$  framework developed in Hennessy (2004), which requires equity market values to construct average  $Q$ . Section 5.1 develops the conceptual framework showing how debt overhang creates a wedge between marginal  $q$  (which governs investment) and average  $Q$  (which is observable), and equation (3) derives the empirical specification that guides our analysis. Without market prices, we would be unable to implement this framework or

interpret our coefficient estimates through the lens of  $Q$ -theory. We clarify this motivation at the beginning of Section 4 with a footnote explaining why the sample is restricted to CRSP-covered firms.

*e. Since tables report  $t$ -stats, it is harder to estimate the range of estimates. At least in the text, the paper should describe the confidence interval for investment effects.*

We are now reporting standard errors in all tables.

*f. It is not obvious that the effects of  $d_i$  should be linear. Especially since less than half of the sample has positive numbers, it would be good to assess where the effects are coming from.*

We address this concern in two ways. First, to ensure that our results are not driven by comparisons between firms with and without long-term liabilities, Column 5 of Table 3 restricts the sample to firms with positive long-term liabilities in 1930. This reduces the sample from 7,074 to 6,048 firm-year observations, focusing on firms that could potentially have gold exposure. The coefficients on  $1933 \times \tilde{d}$  and  $1934 \times \tilde{d}$  are  $-0.058$  and  $-0.043$ , respectively, both statistically significant and similar in magnitude to the baseline estimates.

Second, we conduct additional tests regressing investment on dummies for positive  $\tilde{d}$ , above-median  $\tilde{d}$ , and above-75th percentile  $\tilde{d}$ , with results reported in Internet Appendix Table IA.16. In this design, we interact these  $\tilde{d}$  indicators with four time dummies: 1926–1932 (Before), 1933, 1934, and 1935–1940 (After), using Before as the omitted category.

Column 1 shows that the results hold at the extensive margin: firms with  $\tilde{d} > 0$  invested significantly less than  $\tilde{d} = 0$  firms in 1933 and 1934 relative to the pre-period. The coefficients on  $1933 \times (\tilde{d} > 0)$  and  $1934 \times (\tilde{d} > 0)$  are  $-0.045$  and  $-0.029$ , respectively, both statistically significant at the 1% level. Columns 2 and 3 progressively add interactions for above-median and above-75th percentile exposure. While we observe negative point estimates for high- $\tilde{d}$  interactions—suggesting that more exposed firms may have experienced larger effects—these coefficients are imprecisely estimated. For instance, the coefficient on  $1934 \times (\tilde{d} > \tilde{d}_{0.75})$  is

−0.055, economically substantial but not statistically significant. We interpret this evidence as consistent with non-linear effects but note that the limited variation in  $\tilde{d}$  within the exposed subsample makes it difficult to precisely identify intensive-margin non-linearities. We briefly reference these results in Section 5.5.

*g. Are the effects equally salient for small or large firms? Debt overhang may be worse for firms with fewer alternative sources of financing. If so, how does this affect the aggregate effect calculation?*

Thank you for recommending this test. Internet Appendix Table IA.13 reports results from specifications where the regressors are interacted with firm-level dummies for small firms (below-median assets in 1930), low cash holdings, and high book leverage. Column 1 focuses on firm size: the coefficients on  $\text{Small} \times 1933 \times \tilde{d}$  and  $\text{Small} \times 1934 \times \tilde{d}$  are both negative (−0.049 and −0.073), suggesting that the effect of gold exposure on investment is larger for small firms. The triple interaction is statistically significant at the 5% level for 1934, indicating that small firms experienced notably larger investment declines from gold exposure that year. The results for cash and leverage (columns 2–3) show similar patterns, though the triple interactions are less precisely estimated.

We incorporate these heterogeneous effects into our aggregation exercise. Table 7 reports alternative aggregation results using coefficients from regressions that include triple interaction terms for Low rating (Table 5) and Small (Table IA.13). As you conjectured, incorporating firm size heterogeneity reduces the aggregate gold clause effect in 1934 (from −1.23% to −0.45%), reflecting that large firms—which account for a disproportionate share of aggregate capital—experienced smaller effects. By contrast, the 1933 effect is similar when accounting for size heterogeneity (−1.54% versus −1.79%). We discuss these results in Section 5.6.

*h. The paper states that low ratings have no effect on investment, but it actually seems that the coefficient is too noisily estimated to be conclusive one way or the other. This is important because the paper seems to lever the difference in the ratings effect for investment and payouts.*

You are right that we should not over-interpret the lack of statistical significance. We believe the new results in the revised manuscript, which are based on the extended 1926–1940 dataset and the year-by-year specification (rather than the earlier two-year grouping), provide a more transparent perspective on this issue.

In the revised manuscript, we report results for interactions with low rating in Table 5 and discuss them in Section 5.4. The results show that the coefficient on  $\text{Low} \times 1933$  is negative and statistically significant ( $-0.038$ , significant at the 5% level), indicating that low-rated firms with median gold exposure reduced investment by an additional 3.8 percentage points relative to high-rated firms in 1933. The coefficient for 1934 is positive but only marginally significant. Table 5 also shows that higher  $\tilde{d}$  is associated with higher cash dividends in both 1933 and 1934. Low-rated firms at median gold exposure increased dividends more than high-rated firms in both years, consistent with stronger debt overhang effects for firms closer to default. However, the triple interaction coefficients—capturing the differential marginal effect of gold exposure for low-rated firms—are imprecisely estimated for investment and negative for dividends, suggesting a more nuanced pattern that we discuss in the manuscript.

Importantly, we have toned down the language throughout Section 5.4 to reflect the mixed nature of the evidence. For example, regarding the 1934 investment coefficient, we note: “The positive coefficient in 1934, however, is inconsistent with this prediction, though we note that it is only marginally statistically significant.” For the triple interaction coefficients, we state: “However, the coefficients are imprecisely estimated, so we interpret this evidence with caution.” The section concludes: “In sum, the credit rating results provide mixed but generally supportive evidence for the hypothesis that debt overhang effects are more pronounced for firms closer to default... We therefore interpret these findings as suggestive rather than definitive evidence on the role of credit ratings in mediating debt overhang effects.”

*i. In Table 2 it would be nice to add the number of firms for each column. In fact, I could not find a description of the number of firms with non-zero di.*

We now provide more detailed information on sample composition. Internet Appendix Table IA.3 reports summary statistics separately for  $\tilde{d} > 0$  and  $\tilde{d} = 0$  firms across the three sample periods. Internet Appendix Tables IA.4 and IA.5 provide additional distributional statistics separately for  $\tilde{d} > 0$  and  $\tilde{d} = 0$  firms, including the number of firm-year observations in each group. In the pre-1933 period, we have 1,430 firm-year observations with  $\tilde{d} > 0$  (330 unique firms) and 1,759 firm-year observations with  $\tilde{d} = 0$  (410 unique firms). Tables IA.6 and IA.7 further decompose the  $\tilde{d} > 0$  sample into below-median and above-median exposure groups, showing that high- $\tilde{d}$  firms are slightly smaller and have lower leverage relative to low- $\tilde{d}$  firms within the exposed subsample.

*j. Table 10 states that in Column (3) firms in regulated industries (transportation, communication, and utilities) are excluded. Surely it is strange that these firms were included in the industrials manual in the first place, so they should be dropped everywhere. In fact, these cannot be representative of the industries, which were covered in other manuals.*

You are right, and we apologize for the confusion in the original submission, which was poorly worded on this point. To clarify: transportation, communication, and utilities firms are *not* included in the Moody’s Industrial Manuals—these sectors have their own separate Moody’s manuals (e.g., Moody’s Public Utility Manual, Moody’s Transportation Manual, Moody’s Financial Manual). Our sample is drawn exclusively from the Moody’s Industrial Manuals precisely because industrial firms were not subject to the heavy regulation that characterized utilities, transportation, and communication firms during this period. The original wording incorrectly suggested we were excluding regulated firms from the Industrial Manuals, when in fact we restricted our data collection to the Industrial Manuals to focus on unregulated firms from the outset. We have clarified this in Section 4, which now states that “regulated industries (transportation, communication, and utilities) and financial firms—which are covered in separate Moody’s manuals—are not part of our sample.”

*Additional comment*

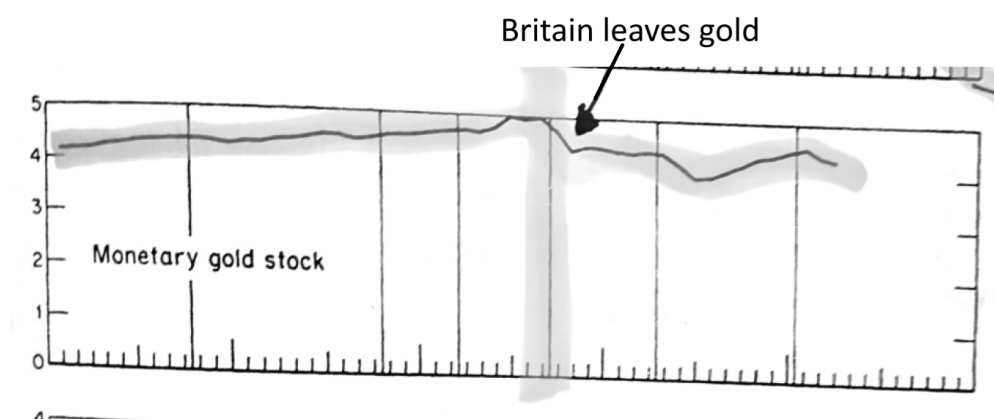


*“They should do a panel, show no pre trends, show that there is no endogenous adjustment in di (and if none, am at a loss as to why good managers in unconstrained firms did not adjust after September 1931.)”*

Thank you for raising this interesting point. We found ourselves grappling with the same puzzle as we conducted the empirical analysis for this revision. As documented in our response to Comment 3, the return evidence suggests that the market initially perceived Britain’s departure as unfavorable to firms with gold-denominated debt. Yet managers did not respond by redeeming their gold bonds in greater numbers prior to the abrogation.

Providing a definitive answer to this puzzle is challenging. However, by studying US politics and monetary policy during this period, we can offer some perspective. The swift policy response of the Fed following Britain’s departure and the policy stances of Presidents Hoover and Roosevelt up to April 1933 can shed some light on the issue. We discuss the chronology of the events below.

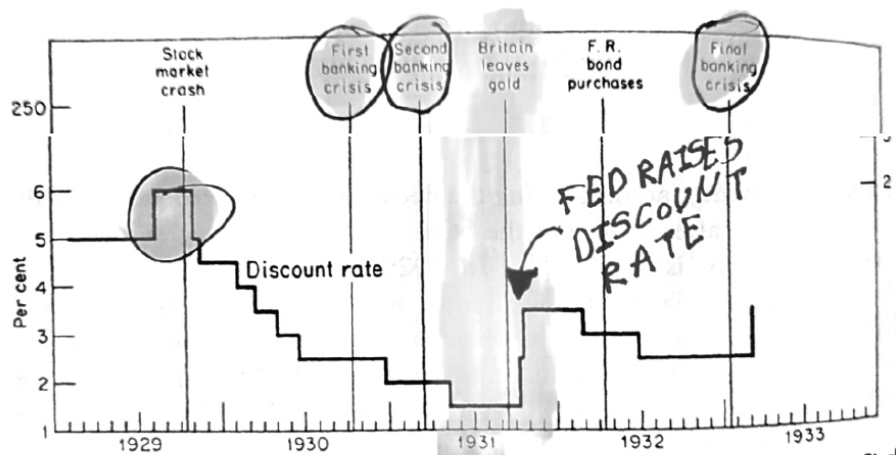
According to Friedman and Schwartz (1963), between September 16 and October 28, speculators (central banks and private holders) converted a substantial amount of their dollar assets to gold anticipating that the US would follow the same path as Great Britain. The joint effect of their action can be seen in Figure R.III.



**Figure R.III.** US gold stock

The Federal Reserve System reacted promptly to the drain. On October 9, the bank raised the discount rate to 2.5% and on October 16, to 3.5%. Figure R.IV shows the sharp increase in the discount rate in the aftermath of Britain’s departure. Following the Fed’s move, the gold outflow

was contained and the gold stock reached its trough at the end of October.



**Figure R.IV.** FED discount rate

According to Friedman and Schwartz (1963), the vigor and promptness of the Fed's actions was unprecedented in its history. The rise in the discount rate was the sharpest and steepest in the history of the Fed, and it arguably discouraged speculators from running on US gold. It is thus conceivable that the decisiveness of the Fed's actions was also enough to persuade managers that the monetary authority was seriously committed to the standard.

These events unfolded within a political context where President Hoover remained a steadfast advocate of the gold standard, which he famously described as "little short of a sacred formula" (Eichengreen and Temin (2000)). In multiple speeches, he asserted that the United States possessed both the ability and the moral obligation to maintain the gold standard.

During the 1932 presidential campaign, future president Roosevelt himself emphatically denied that he would alter the value of gold in response to Hoover's claim to the contrary. As late as March 6, 1933, the secretary of the Treasury, Will Woodin, had even reiterated the administration's commitment to the gold standard saying that it was "ridiculous and misleading to say that we are off the gold standard.... We are definitely on the gold standard." Thus, it seems conceivable that the strong bipartisan support the gold standard received up to April 1933 was sufficient to convince firm managers that going off gold wasn't a serious possibility.

Finally, there is also the more mundane aspect of the cost involved in redeeming a firm's outstanding corporate bonds. In those years, bond prices were severely depressed. Figure R.V reports quotes (% of par value) for several industrial corporate bonds at the end of September 1931.

| Industrial and Miscellaneous Securities                                                                                                                                                                                                                                                                                                                                                                                                      |             |      |                                   |      |      |                                   |      |      |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------|-----------------------------------|------|------|-----------------------------------|------|------|--|
| Under the heading "Industrial and Miscellaneous Securities" we include all issues which do not appear under the previous two headings, namely "Railroads (Steam)" and "Public Utilities." In the case of stocks, however, we put "Textile Manufacturing," "Insurance," "Mining," "Real Estate and Land," "Title Guarantee and Safe Deposit Companies" under separate heads, then follow with the rest of the "Industrial and Miscellaneous." |             |      |                                   |      |      |                                   |      |      |  |
| NOTICE.—All bond prices are "and interest" except where marked "I" and income and defaulted bonds.                                                                                                                                                                                                                                                                                                                                           |             |      |                                   |      |      |                                   |      |      |  |
| Bonds.                                                                                                                                                                                                                                                                                                                                                                                                                                       | Bid.        | Ask. | Bonds.                            | Bid. | Ask. | Bonds.                            | Bid. | Ask. |  |
| <b>INDUSTRIAL AND MISCELLANEOUS BONDS.</b>                                                                                                                                                                                                                                                                                                                                                                                                   |             |      | Beech Creek C & O 5s '44. J&D     | 94½  | 100  | Crown Willamet Pap 6s '51. J&J    | 80   | 83   |  |
| Abbots Dairies deb 6s 1942. M&B                                                                                                                                                                                                                                                                                                                                                                                                              | ---         | 99½  | Beld's-Hem'way 6½ notes '36 J&J   | ---  | ---  | Crown Zellerbach Corp—            | ---  | ---  |  |
| Abtibi Pow & Paper—                                                                                                                                                                                                                                                                                                                                                                                                                          | ---         | ---  | Belgo-Can Pap 1st 6s 1943. J&J    | ---  | ---  | Deb 6s 1940 with warr. M&B        | 56   | 58   |  |
| 1st M 5s 1953 ser A. J&D                                                                                                                                                                                                                                                                                                                                                                                                                     | 39          | 39½  | Beneficial Indus Loan Corp—       | ---  | ---  | Crucible Steel Co deb 5s '40. M&N | ---  | 88   |  |
| Abraham & Straus Inc—                                                                                                                                                                                                                                                                                                                                                                                                                        | ---         | ---  | Conv deb 6s 1946. M&S             | 80   | 81   | Cuba Cane Products—               | ---  | ---  |  |
| Deb 5½s 1943 with warr. A&O                                                                                                                                                                                                                                                                                                                                                                                                                  | 92          | 94   | Beth'm St 1st 1 & ref 5s '42. M&N | 98   | 100  | Deb 6s 1950. J&J                  | 4½   | 4½   |  |
| Adams Exp coll tr g 4s '48. M&B                                                                                                                                                                                                                                                                                                                                                                                                              | 70          | 84   | Purch money 5s 1936. J&J          | 100½ | 101½ | Cuba Cane Sug deb 7s 1930. J&J    | ---  | ---  |  |
| Coll tr g 4s 1947. J&D                                                                                                                                                                                                                                                                                                                                                                                                                       | 75          | 79   | Penn-Mary Steel 5s 1937. J&J      | 97   | ---  | Conv deb 5s 1930. J&J             | ---  | ---  |  |
| Alax Rubber s f 8s 1936. J&D                                                                                                                                                                                                                                                                                                                                                                                                                 | ---         | 14   | Bluff Point Land Impt Co—         | ---  | ---  | Cuban Dominican Sugar Co—         | ---  | ---  |  |
| Alabama Cons Coal & Iron—                                                                                                                                                                                                                                                                                                                                                                                                                    | ---         | ---  | 1st mtge guar 4s 1940. J&J        | 80   | 90   | 1st lien s f 7½s 1944. M&N        | 5    | 11½  |  |
| 1st cons M 5s 1933. M&N                                                                                                                                                                                                                                                                                                                                                                                                                      | 75          | 90   | Bohemian (First) Glass Works—     | ---  | ---  | Certificates of deposit.---       | ---  | ---  |  |
| Alabama Mills 1st 6½s '43 A&O                                                                                                                                                                                                                                                                                                                                                                                                                | 5           | 10   | 1st 7s 1957 without warr. J&J     | ---  | ---  | Stmpd with stk purch warr.---     | 6    | 7    |  |
| Ala Steel & Shipbldg—See Tenn                                                                                                                                                                                                                                                                                                                                                                                                                | O. I. & RR. | 10   | Borden Mills 1st s f 6s 1934. F&A | 85   | 95   | Certificates of deposit.---       | ---  | ---  |  |
| Alaska Gold Mines deb 6s 25M&B                                                                                                                                                                                                                                                                                                                                                                                                               | 5½          | 10   | Boston Store (Ohio) 5s 1938.---   | 100½ | 101½ | Oudahy Pack s f 5s 1946. J&D      | 101½ | 102  |  |
| Deb 6s 1926 ser B. M&B                                                                                                                                                                                                                                                                                                                                                                                                                       | 5½          | 10   | Boston Term Co 3¼s 1947. F&A      | 92½  | 94   | Sink fund deb g 5½s 1937 A&O      | 90½  | 92   |  |
| Albany Perf Wrapping Corp—                                                                                                                                                                                                                                                                                                                                                                                                                   | ---         | ---  | Botany Consol Mills 6¼s '34 A&O   | 20½  | 22   | Ouyamel Fruit 1st 6s 1940. A&O    | ---  | 103½ |  |
| 1st M coll tr 6s 1948. A&O                                                                                                                                                                                                                                                                                                                                                                                                                   | 50½         | 58½  | Brown Co 1st 5¼s 1946. A&O        | 63   | 65   | Dairymen's League Co-oper Assn    | ---  | ---  |  |
| Alberta Pacific Grain 6s 1946.---                                                                                                                                                                                                                                                                                                                                                                                                            | ---         | ---  | 1st 5¼s 1950 ser B. M&S           | 63   | 65   | 6s 1939 ser CO.---                | ---  | ---  |  |
| Alkoma Steel 1st 5s 1962. A&O                                                                                                                                                                                                                                                                                                                                                                                                                | 30          | 32   | Buffalo & Susq Iron 5s 1932 J&D   | 96   | 98   | 6s 1937 ser DD.---                | ---  | ---  |  |
| Allied Packers deb 6s 1939. J&J                                                                                                                                                                                                                                                                                                                                                                                                              | 30½         | ---  | Burneister & Wain (Copenhagen)    | ---  | ---  | De Bardeleben Coal Corp—          | ---  | ---  |  |
| 1st M & coll tr 8s 1939. J&J                                                                                                                                                                                                                                                                                                                                                                                                                 | 30½         | ---  | 15-yr s f extl 6s 1940. J&J       | ---  | ---  | 1st mtge 6s 1953. J&D             | 39   | 40   |  |
| Allis-Chalmers Mfg Co—                                                                                                                                                                                                                                                                                                                                                                                                                       | ---         | ---  | Bush Term 1st 4s 1952. A&O        | 91½  | ---  | Dery (D G) 7s 1942 stpd. M&B      | 1    | 20   |  |
| Deb gold 5s 1937. M&N                                                                                                                                                                                                                                                                                                                                                                                                                        | ---         | 96½  | Cons g 5s Jan 1955. J&J           | 82   | 85   | Second stamped.---                | 2½   | 64   |  |
| Alpine Montan Steel Corp—                                                                                                                                                                                                                                                                                                                                                                                                                    | ---         | ---  | Bush Term Bldgs 1st 5s '60. A&O   | ---  | 95½  | Denver Un StkYds 5s 1946. J&J     | 99   | 101  |  |
| 1st s f 7s 1955. M&S                                                                                                                                                                                                                                                                                                                                                                                                                         | 70          | 70   | By-Products Coke Corp—            | 50   | 83   | Deutsche Bank (Berlin)---         | 82½  | 83½  |  |
| Aluminum Co deb 5s 1952. M&S                                                                                                                                                                                                                                                                                                                                                                                                                 | 98½         | 99½  | 1st M 5¼s 1945 ser A. M&N         | 3    | 6    | 6½ note '32 (Am part cts) M&B     | 43   | 46   |  |
| Aluminum Ltd s f debt 5s '48. J&J                                                                                                                                                                                                                                                                                                                                                                                                            | 85          | 88   | Cady Lumber 6¼s 1939. M&N         | ---  | ---  | Deutsche Rentenbank—              | ---  | ---  |  |
| Amalgamated Sugar Co—                                                                                                                                                                                                                                                                                                                                                                                                                        | ---         | ---  | California Packing Corp—          | ---  | ---  | Farm Lns s f 6s Oct 15 '60 A&O    | 83½  | 84   |  |
| 1st s f 7s 1937. A&O                                                                                                                                                                                                                                                                                                                                                                                                                         | 60          | 75   | Cons deb 5s 1940. J&J             | 82   | 83   | Dodge Bros deb 6s 1940. M&N       | ---  | ---  |  |
| Amer Aggregates Corp—                                                                                                                                                                                                                                                                                                                                                                                                                        | ---         | ---  | Calif Pet deb 6¼s 1938. M&N       | 89   | 90   | Dodge Mfg 1st 7s 1942. J&J        | 50   | 53½  |  |
| Deb 6s '43 ser A with war. F&A                                                                                                                                                                                                                                                                                                                                                                                                               | 60½         | 65   | Conv s f deb 6s 1939. F&A         | 79   | ---  | Bold (Jac) Pack 1st 6s '42. M&N   | ---  | ---  |  |
| Without warrants.---                                                                                                                                                                                                                                                                                                                                                                                                                         | 58          | ---  | Camaguey Sugar 7s 1942 A&O        | 17   | 20   | Dominion Cannera 1st 6s '40 A&O   | 84   | 88   |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                              |             |      | Can Cement 1st 5¼s 1947. M&N      | ---  | ---  |                                   |      |      |  |

Figure R.V. Industrial bond quotes. Commercial and Financial Chronicle, Oct 1931

As the figure shows, many of the bonds were traded at a steep discount with respect to their par value of 100. Redeeming the bonds by exercising call options, to the extent that the bond had one, would have been quite costly. The strike price of those options is typically at a premium above par. Purchasing the bonds through open market operations might also have been a costly option given the price impact that a large purchase would likely involve.

For these reasons, while early redemption might have been optimal in the absence of frictions and with perfect foresight, it is plausible that most managers either did not anticipate the abrogation or deemed early redemption too costly.

Thank you again for your constructive and insightful comments. The new analyses, guided

by your suggestions, have significantly strengthened the empirical evidence and improved our understanding of the mechanism, providing a more comprehensive perspective on this period. In particular, your questions prompted us to engage more deeply with the political and economic context of the time, from which we have learned a great deal.

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